FIITJEE Students From All Programs have bagged 33 in Top 100, 77 in Top 200 and 205 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2015: 4551 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2015.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE (Main), 2016

HALF COURSE TEST – VII

Time Allotted: 3 Hours **Maximum Marks: 360**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 30)** contains 30 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19} C$
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12} C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5 N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

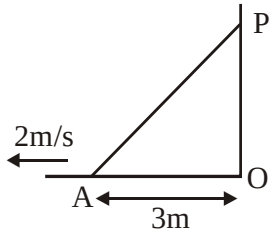
PART – I

SECTION – A

(Single Correct Choice Type)

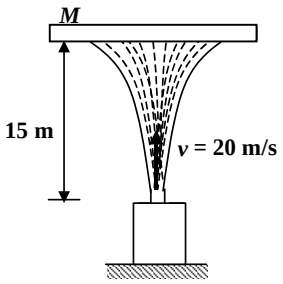
This section contains ***30 multiple choice questions***. Each question has four choice (A), (B), (C) and (D) out of which ***ONLY ONE*** is correct.

1. The end A of a ladder AP of length 5 m, kept inclined to a vertical wall is slipping over a horizontal surface with velocity of 2 m/s, when A is at a distance of 3m from the wall. Velocity of centre of mass at this moment is



(A) 1.25 m/s (B) 0 m/s
(C) 1 m/s (D) 2 m/s

2. A vertical jet of water coming out of a nozzle with velocity 20 m/s supports a plate of mass *M* stationary at a height *h* = 15m, as shown in the figure. If the rate of water flow is 1 litre per second, the mass of the plate is (Assume the collision to be *inelastic*).



(A) 1 kg (B) 1.414 kg
(C) 2 kg (D) 10 kg

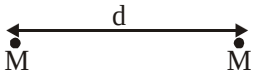
3. The binding energy of a particle of mass m with a planet, when it is on the planets surface, is $\frac{1}{2}mv_0^2$. A tunnel is dug along a diameter of the planet and the particle is dropped into it from the surface, when the body reaches the centre of the planet, its speed is

(A) v_0 (B) $\frac{v_0}{\sqrt{2}}$

(C) Zero (D) $\frac{v_0}{2}$

Space for rough work

4. Consider the two identical particles shown in the given figure. They are released from rest and can move towards each other under the influence of their mutual gravitation forces. Speed of each particle, when the separation reduces to half of initial value equals



(A) $\sqrt{\frac{GM}{d}}$ (B) $\sqrt{\frac{2GM}{d}}$

(C) $\sqrt{\frac{GM}{2d}}$ (D) none of these

5. A gas of adiabatic exponent γ is supplied heat at a constant pressure. What is the ratio between dQ, dW and dU in the process

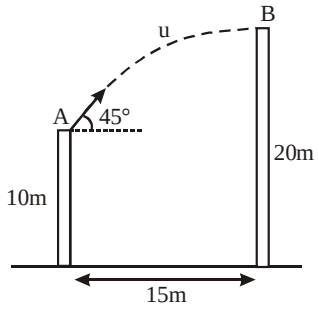
(A) $\gamma : \gamma - 1 : \frac{1}{\gamma}$ (B) 1 : 1 : $\gamma - 1$

(C) $\gamma : \gamma - 1 : 1$ (D) $\gamma : 1 : \gamma - 1$

6. Find the value of 'u' so that the ball reaches at point B.
(Take g = 10 m/s^2)

(A) 20 m/s (B) 40 m/s

(C) 15$\sqrt{2}$ m/s (D) 50 m/s



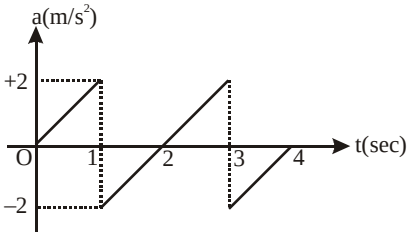
7. Acceleration vs time graph is shown in the figure for a particle moving along a straight line. The particle is initially at rest. Find the time instant(s) when the particle comes to rest ?

(A) t = 0, 1, 2, 3, 4

(B) t = 0, 2, 4

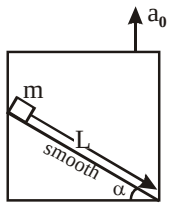
(C) t = 1, 3

(D) None of these



Space for rough work

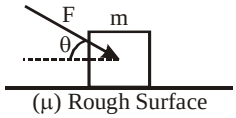
8. A body of mass 'm' is released from rest on an inclined plane in an elevator (moving up with acceleration a_0). Find the time taken by the body to reach the bottom of the inclined plane.



(A) $\sqrt{\frac{L}{(a_0+g)\sin\alpha}}$ (B) $\sqrt{\frac{2L}{(a_0+g)\sin\alpha}}$

(C) $\sqrt{\frac{3L}{(a_0+g)\sin\alpha}}$ (D) $2\sqrt{\frac{L}{(a_0+g)\sin\alpha}}$

9. Find minimum value of the angle θ so that block of mass m does not move on rough surface. The coefficient of static friction between the block and surface is μ whatever may be the value of applied force F.



(A) $\tan^{-1}(\mu)$ (B) $\frac{1}{2}\tan^{-1}(\mu)$

(C) $\cot^{-1}(\mu)$ (D) $\frac{1}{2}\cot^{-1}(\mu)$

10. A steel ring of radius r and cross-sectional area A is fitted on to a wooden disc of radius R(R > r). If Young's modulus be Y, then the force with which the steel ring is expanded is :

(A) $AY\frac{R}{r}$ (B) $AY\left(\frac{R-r}{r}\right)$

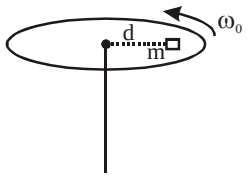
(C) $\frac{Y}{A}\left(\frac{R-r}{r}\right)$ (D) $\frac{Yr}{AR}$

11. A rod of length 40 cm has the coefficient of linear expansion $\alpha_1 = 6 \times 10^{-6}/^oC$. Another rod of length *l* has the coefficient of linear expansion $\alpha_2 = 4 \times 10^{-6}/^oC$. If the difference in the lengths of the two rods always remain same at all temperatures, then the value of *l* is

(A) 26 cm (B) 60 cm

(C) 80 cm (D) 32 cm

12. A small block of mass 'm' is placed on a rough rotating table. Find maximum angular velocity (ω_0) that can be given to the table so that the particle does not slip on the table. It is given that the coefficient of friction between the block and the table is (μ).



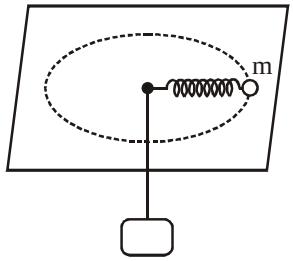
(A) $\sqrt{\frac{2\mu g}{d}}$ (B) $\sqrt{\frac{3\mu g}{d}}$

(C) $\sqrt{\frac{\mu g}{d}}$ (D) $2\sqrt{\frac{\mu g}{d}}$

Space for rough work

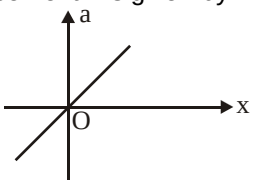
13. A body executes simple harmonic motion. The potential energy (PE), kinetic energy (KE) and total energy (TE) are measured as a function of displacement x. Which of the following statements is true?
(A) TE is zero when x = 0 (B) PE is maximum when x = 0
(C) KE is maximum when x = 0 (D) KE is maximum when x is maximum

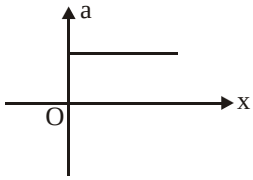
14. A mass m on a frictionless table is attached to a hanging mass by a spring (of spring constant K and natural length ℓ_0) and a cord through a hole in the table. If the hanging mass remains stationary and the mass m moves on a circular path with angular velocity ω the radius of its circular path is

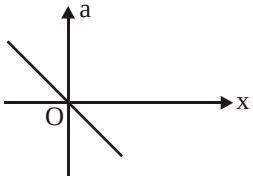


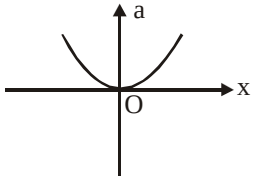
(A) ℓ_0 (B) $\frac{m\omega^2\ell_0}{K-m\omega^2}$
(C) $\frac{K\ell_0}{(K-m\omega^2)}$ (D) $\frac{K\ell_0}{K+m\omega^2}$

15. A progressive wave is travelling in a medium such that frequency of oscillation and displacement amplitude of the particles of the medium are f and A respectively. The ratio of their acceleration amplitude and velocity amplitude is
(A) $2\pi f$ (B) πf
(C) $\frac{2\pi f}{A}$ (D) $\frac{\pi f}{A}$.

16. For a particle executing simple harmonic motion, the correct variation of acceleration a with displacement x is given by
(A)


(B)


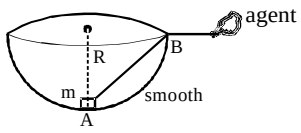
(C)


(D)


Space for rough work

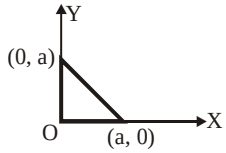
17. An open organ pipe of length 'L' vibrates in second harmonic mode. The pressure variation is maximum, (neglect end corrections)
(A) at the two ends
(B) at a distance $\frac{L}{4}$ from either end inside the tube
(C) at the mid–point of the tube
(D) none of these.

18. The minimum work done by the agent, in pulling a small particle of mass m from A to B as shown in figure, is



(A) 4 mgR (B) mgR
(C) 3mgR (D) 2mgR

19. Three rods of the same mass are placed as shown in the figure. What will be the co-ordinates of the center of mass of the system?



(A) (a/2, a/2) (B) $\left(\frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}}\right)$
(C) $\left(\frac{\sqrt{2}a}{3}, \frac{\sqrt{2}a}{3}\right)$ (D) $\left(\frac{a}{3}, \frac{a}{3}\right)$

20. The density ρ of a liquid varies with depth h from the free surface as $\rho = kh$. A small body of density ρ_1 is released from the surface of liquid. The body will
(A) come to a momentary rest at a depth $\frac{2\rho_1}{k}$ from the free surface
(B) execute simple harmonic motion about a point at a depth $\frac{\rho_1}{k}$ from the surface
(C) execute simple harmonic motion of amplitude $\frac{\rho_1}{k}$
(D) all of the above

Space for rough work

21. At the center of a non-uniform ring of radius R, made up of two uniform halves of mass 2M and M (G : Newton's gravitational constant)
(A) field and potential both are zero
(B) field is zero but potential is $\frac{-3GM}{R}$
(C) field is zero but potential is $\frac{-GM}{R}$
(D) magnitude of field is $\frac{2GM}{\pi R^2}$ and potential is $-\frac{3GM}{R}$

22. For a satellite of mass m orbiting the earth very close to earth's surface (mass of earth = M, radius of earth = R) total energy is
(A) zero
(B) greater than zero
(C) $\frac{GMm}{R}$
(D) $-\frac{GMm}{2R}$

23. An ideal gas has initial volume V and pressure P. In doubling its volume the minimum work done will be in the following process (of given processes)
(A) Isobaric process
(B) Isothermal process
(C) Adiabatic process
(D) None of the above

24. A particle moving in the positive x-direction has initial velocity v_0. The particle undergoes retardation kv^2, where v is its instantaneous velocity. The velocity of the particle as a function of time is given by
(A) $v = v_0/(1 + kv_0t)$
(B) $v = \frac{2v_0}{1+kt}$
(C) $v = \frac{v_0}{kt}$
(D) $v = \frac{v_0}{(1+k^2v_0^2t)}$

25. Velocity of a particle at any instant is given by the equation $\vec{v} = \left(2t\hat{i} + 3t^2\hat{j}\right)$ m/s and radius of the curvature of the path is 2m. Centripetal acceleration of the particle at t = 2 s is
(A) 80 m/s^2
(B) 160 m/s^2
(C) 40 m/s^2
(D) 100 m/s^2

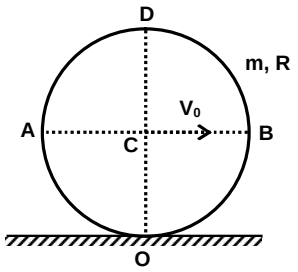
Space for rough work

26. The elongation of a freely hanging uniform steel rope, if its length is doubled, will increase in the ratio of

(A) 2 : 1 (B) 4 : 1

(C) 8 : 1 (D) 16 : 1

27. A uniform ring of mass m and radius R is doing uniform pure rolling motion on a horizontal surface. The velocity of the centre of the ring is V_0. The kinetic energy of the segment AOB is



(A) $\frac{mV_0^2}{2}-\frac{mV_0^2}{\pi}$ (B) $\frac{mV_0^2}{2}+\frac{mV_0^2}{\pi}$

(C) $\frac{mV_0^2}{2}$ (D) mV_0^2.

28. A rope of mass 'm' is looped in a circle of radius R and rotated with a constant angular velocity ω_0 about its axis in gravity free space. Find the tension in the rope ?

(A) $T = mR\omega_0^2$ (B) 2 m R $\pi\omega^2$

(C) $T = \frac{mR\omega_0^2}{2\pi}$ (D) $4mR\pi\omega_0^2$

29. The displacement function of a oscillating body is given by x = 0.3 $\sin\left(10\pi t+\frac{\pi}{6}\right)$ where x and t are measured in meter and second respectively

(A) the period of oscillation is 5 s

(B) the body starts its motion from the equilibrium

(C) the minimum time the body takes to reach the equilibrium is 1/12 sec

(D) all of these

30. A simple pendulum is attached to the roof of a stationary elevator. If the time period of oscillation is T when it is stationary, what will be time period of oscillation when the elevator falls freely

(A) 0 (B) T/2

(C) 2T (D) none of these

Space for rough work

Chemistry

PART – II

SECTION – A

(Single Correct Choice Type)

This section contains ***30 multiple choice questions***. Each question has four choice (A), (B), (C) and (D) out of which ***ONLY ONE*** is correct.

1. Which of the following is correct about the reaction?

   $3NaClO \xrightarrow{\Delta} NaClO_3 + 2NaCl$

   (A) It is disproportionation reaction
   (B) Oxidation number of chlorine decreases as well as increases in this reaction
   (C) This reaction is used for the manufacturing of halates
   (D) All the above

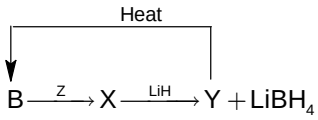
2. $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

   For the above equilibrium, which is correct?
   (A) Equilibrium will shift in forward direction by addition He at constant volume
   (B) Equilibrium will shift in forward direction by addition of He at constant pressure
   (C) Equilibrium with shift in backward direction by addition of He at constant volume
   (D) Equilibrium will shift in backward by addition of He at constant pressure

3. In the reaction $A_2(g) + 4B_2(g) \rightleftharpoons 2AB_4(g)$, $\Delta H < 0$. The decomposition of $AB_4(g)$ will be not favored at
   (i) Low temperature and high pressure
   (ii) High temperature and low pressure
   (iii) Addition of inert gas at constant pressure
   (iv) Addition of inert gas at constant volume
   (v) Using catalyst
   (vi) Compressing reaction mixture by mass less and friction less piston
   (A) (ii), (iii) (B) (i), (iv), (v), (vi)
   (C) (i), (ii), (iii), (iv) (D) (ii), (iii), (iv), (v), (vi)

4. Diborane reacts with ammonia under different condition to give a variety of products. Which one of among the following is not formed in these conditions?
   (A) $B_2H_6.2NH_3$ (B) $B_{12}H_{12}$
   (C) $B_3N_3H_6$ (D) $(BN)_3$

Space for rough work

5. 

Which of the statement is true for the above sequence of reactions?

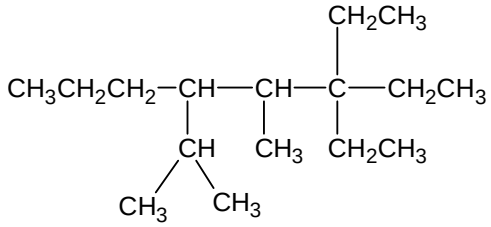
(A) Z is hydrogen (B) X is B_2H_6

(C) A and Y are F_2 and B_2H_6 respectively (D) Z is potassium hydroxide

6. Find the value of wave number $\bar{\nu}$ in turns of Rydberg's constant, when transition of electron takes place between two level of He^+ ion whose sum is 4 and difference is 2.

(A) 8R/9 (B) 32R/9

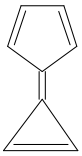
(C) 3R/4 (D) none

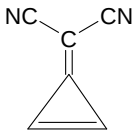
7. The IUPAC name of the compund 

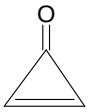
(A) 3, 3 –diethyl-4-methyl-5-isopropyloctane (B) 3, 3 –diethyl-5-methyl-5-isopropyloctane

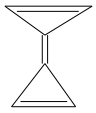
(C) 4-isopropyl-5-metheyl-6, 6-diethyloctane (D) 6, 6-diethyl-4-isopropyl-5-metheyloctane

8. Which of the following is non - aromatic?

(A) 

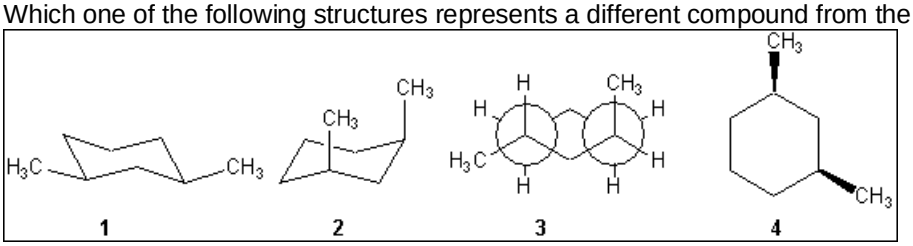
(B) 

(C) 

(D) 

Space for rough work

9. Which one of the following structures represents a different compound from the other three?



(A) 1 (B) 2
(C) 4 (D) 3

10. About the compound 3, 3–diethyl-4-methyl-5-isopropyloctane which of the following statement is/are not true
(A) It has three 3^0 carbon (B) It has one 4^0 carbon
(C It has six 1^0 carbon (D) It has ten 2^0 hydrogen

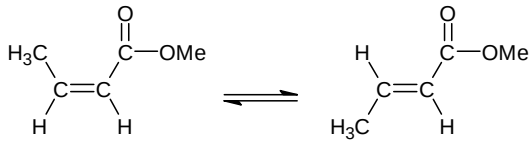
11. When a substance A reacts with water it produces a combustible gas B and a solution of a substance C in water. When another substance D reacts with this solution of C, it also produces the same gas B even on reaction with dilute sulphuric acid at room temperature. A imparts a deep golden yellow colour to a smokeless flame of Bunsen burner. A, B, C and D respectively are
(A) Na, H_2, NaOH and Zn (B) K, H_2, KOH and Al
(C) Ca, H_2, $Ca(OH)_2$ and Sn (D) CaC_2, C_2H_2, $Ca(OH)_2$ and Fe

12. Which of the following statements is incorrect?
(A) The hydrolysis of $(CH_3)_3SiCl$ gives a disiloxane.
(B) The hydrolysis of $(CH_3)_2SiCl_2$ gives a chain compound.
(C) The hydrolysis of CH_3SiCl_3 gives a cross-linked polymer.
(D) The hydrolysis of $(CH_3)_2SiCl_2$ gives $(CH_3)_2SiO_2$.

13. A compound X on heating gives a colourless gas. The residue is dissolved in water to obtain Y. Excess CO_2 is bubbled through aqueous solution of Y when Z is formed. Z on gentle heating gives back X. The X is
(A) $CaCO_3$ (B) $Ca(HCO_3)_2$
(C) Na_2CO_3 (D) $NaHCO_3$

Space for rough work

14. Consider the following processes. The correct prediction regarding the above isomerization is/are:



(A) $K_{eq} > 1$ for backward reaction
(B) Isomerization is due to existence of resonance properties in the molecule
(C) C = C double bond length is shorter than the in a normal alkene.
(D) Electron moves opposite to carbonyl group during resonance.

15. The enolic from of acetone contains
(A) 9 σ bonds, 1π bond and 2 lone pairs
(B) 8 σ bonds, 2π bond and 2 lone pairs
(C) 10σ bonds, 1π bond and 1 lone pairs
(D) 9σ bonds, 2π bond and 1 lone pairs

16. Consider the reaction: $2H_2(g) + 2NO(g) \longrightarrow N_2(g) + 2H_2O(g)$
The rate law for this reaction is
Rate = $K[H_2][NO]^2$
Under what conditions could these steps represent the mechanism?

Step : 1 $\quad 2NO \rightleftharpoons N_2O_2$

Step : 2 $\quad N_2O_2 + H_2 \longrightarrow N_2O + H_2O$

Step : 3 $\quad N_2O + H_2 \longrightarrow N_2 + H_2O$

(A) These steps cannot be the mechanism under any circumstances
(B) These steps could be the mechanism if step 1 is the slow step
(C) These steps could be the mechanism if step 2 is the slow step
(D) These steps could be the mechanism if step 3 is the slow step

17. If $Ca_3(PO_4)_2$ and H_3PO_3 contain same number of 'P' atom, then the ratio of oxygen atoms in these compounds respectively is
(A) 8/3
(B) 2/3
(C) 3
(D) 4/3

Space for rough work

18. Give that for a reaction of nth order, the integrated rate equation:

$K = \frac{1}{(n-1)t}\left[\frac{1}{C^{n-1}} - \frac{1}{C_0^{n-1}}\right]$, where C and C_0 are the concentration of reactant at time t and initially respectively. The $t_{3/4}$ and $t_{1/2}$ are related as

(A) $t_{3/4} = t_{1/2}\left[2^{n-1}+1\right]$ (B) $t_{3/4} = t_{1/2}\left[2^{n-1}-1\right]$

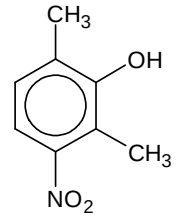
(C) $t_{3/4} = t_{1/2}\left[2^{n+1}+1\right]$ (D) $t_{3/4} = t_{1/2}\left[2^{n+1}-1\right]$

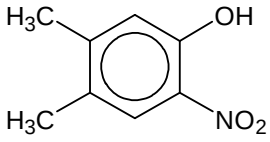
19. In which on of the following species the central atom has the type of hybridization which is not the same as that present in the other species?

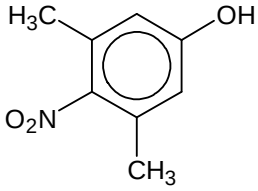
(A) SF_4 (B) I_3^-

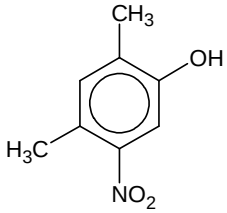
(C) $SbCl_5^{2-}$ (D) PCl_5

20. Which one of the following phenols will show highest acidity?

(A) 

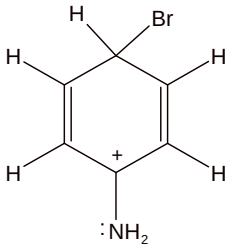
(B) 

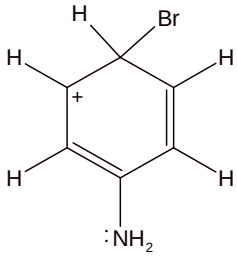
(C) 

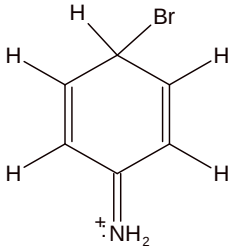
(D) 

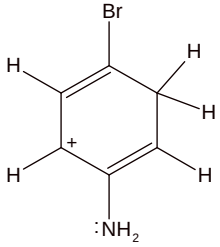
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21. Which one of the following is most stable?

(A) 

(B) 

(C) 

(D) 

22. Rank the following free radical in order of decreasing stability:

(I) $C_6H_5\dot{C}HC_6H_5$ (II) $C_6H_5—\dot{C}H—CH=CH_2$ (III) $CH_3—\dot{C}H—CH_3$

(IV) $C_6\dot{H}_5—CH—CH_3$ (V) $CH_3CH=CHCH_2\dot{C}H_2$ (VI) $CH_3—CH_2—\overset{\dot{}}{\underset{CH_3}{\underset{|}{C}}}—CH_3$

(A) I > II > III > IV > V > VI
(B) VI > V > IV > III > II > I
(C) I > II > IV > VI > III > V
(D) I > IV > VI > V > II > III

23. Consider the flowing reversible system $A(g)+2B(g)\rightleftharpoons AB_2$ $K_c = 1/2$. The above equilibrium is established in a 1L flask and at equilibrium 2 mol of each A and B are present. If 2 mol of B and 2x mol of AB_2 is added further so that mol of A does not change then the value of x is

(A) 6
(B) 8
(C) 3
(D) 4

24. 0.02 M HCOOH has degree of dissociation 10%. The K_a of HCOOH is

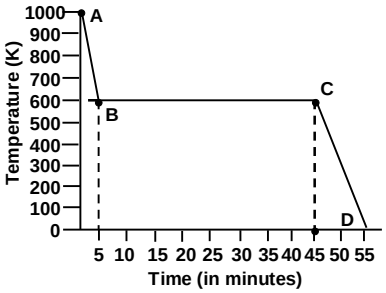
(A) 2×10^{-3}
(B) 2×10^{-4}
(C) 2×10^{-5}
(D) 2×10^{-2}

Space for rough work

25. The degree of dissociation of two weak acids, α_1 and α_2 are in ratio 1 : 2. The value of $K_{a_1} = 2\times10^{-4}$, the value of K_{a_2} is (concentration of both acids are same)

(A) 8×10^{-4} (B) 2×10^{-4}
(C) 4×10^{-4} (D) 1×10^{-4}

26. Two moles of a substance is cooled at the rate of 0.4 kJ min^{-1} as shown in graph. Curve AB, B and C and curve CD represent respectively, the cooling of liquid, start of freezing, completion of freezing and cooling of the solid based on this data. The change in entropy of fusion in J $mol^{-1}K^{-1}$ is



(A) 03.33 (B) 13.33
(C) 23.33 (D) 33.33

27. Two moles of CO and one mole of O_2 are taken in a container of volume one litre to form two moles of CO_2 according to equation:

$2CO(g) + O_2(g) \longrightarrow 2CO_2(g);\ \Delta H = -560$ kJ

The pressure in the vessel changes from 70 atm to 40 atm. If the gases deviate appreciably from ideal nature (1.0 L atm = 0.1 kJ), then

(A) $\Delta U = -555.895$ kJ (B) $\Delta U = -518.95$ kJ
(C) $\Delta U = -557$ kJ (D) $\Delta U = +518.95$ kJ

28. If the solubility of $SrSO_4$ in water, 0.01 M Na_2SO_4 and 0.02 M $Sr_3(PO_4)_2$ be S_1, S_2 and S_3, then select the correct order of solubility

(A) $S_1 < S_2 < S_3$ (B) $S_1 > S_2 > S_3$
(C) $S_1 < S_3 < S_2$ (D) $S_3 < S_1 < S_2$

29. The average speed of an ideal gas molecule at 27°C is 0.3 ms^{-1}. The average speed of the same at 927°C will be

(A) 0.6 ms^{-1} (B) 0.3 ms^{-1}
(C) 0.9 ms^{-1} (D) 3 ms^{-1}

30. Which of the following metal reacts most vigorously with water?

(A) Li (B) Na
(C) K (D) Be

Space for rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

*This section contains **30 multiple choice questions** numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct*

1. If l,m,n are in A.P., then the straight line lx + my + n = 0 will pass through the point
   (A) (–1, 2) (B) (1, – 2)
   (C) (2, 1) (D) (1, 2)

2. Equation of lines which passes through the points of intersection of the lines 4x – 3y – 1 = 0 and 2x – 5y + 3 = 0 and are equally inclined to the axes are
   (A) $y \pm x = 0$ (B) $y - 1 = \pm 1(x - 1)$
   (C) $x - 1 = \pm 2(y - 1)$ (D) none of these.

3. The circle $x^2 + y^2 + 4x - 4y + 4 = 0$ touches
   (A) x-axis (B) y-axis
   (C) both axes (D) none of these.

4. If the line x + 2by + 7 = 0 is a diameter of the circle $x^2 + y^2 - 6x + 2y = 0$, then b =
   (A) 3 (B) – 5
   (C) 5 (D) – 1.

5. The centre of $14x^2 - 4xy + 11y^2 - 44x - 58y + 71 = 0$
   (A) (2, 3) (B) (2, – 3)
   (C) (–2, 3) (D) (–2, – 3)

6. Vertex of the parabola $x^2 + 4x + 2y - 7 = 0$ is
   (A) $\left(-2, \frac{11}{2}\right)$ (B) (–2, 2)
   (C) (–2, 11) (D) (2, 11).

Space for rough work

7. If latus rectum of an ellipse be equal to half of its minor axis then its eccentricity is
(A) rational number (B) irrational number
(C) ellipse not possible (D) Integer

8. If distance between the directices be thrice the distance between the focii then eccentricity of the ellipse is
(A) $\frac{1}{\sqrt{3}}$ (B) $\frac{2}{3}$
(C) $\frac{1}{2}$ (D) $\frac{7}{8}$.

9. By shifting the vertex of a parabola which of the following will definitely change
(A) latus rectum (B) axis of symmetry
(C) directrix (D) focus

10. The distance of the point 'θ' on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ from a focus is
(A) $a(e + \cos\theta)$ (B) $a(e - \cos\theta)$
(C) $a(1 + e\cos\theta)$ (D) $a(1 + 2\cos\theta)$.

11. The ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ and the hyperbola $\frac{x^2}{25}-\frac{y^2}{16}=1$ have in common
(A) centre only (B) centre, foci and directrices
(C) centre, foci and vertices (D) centre and vertices only.

12. If $x=\sqrt{7+4\sqrt{3}}$ then $x+\frac{1}{x}=$
(A) 4 (B) 6
(C) 2 (D) 3.

Space for rough work

13. If l, m, n are real and $l \neq m$, then the roots of the equation $(l - m)x^2 - 5(l + m)x - 2(l - m) = 0$ are
(A) complex (B) real & distinct
(C) real and equal (D) none of these.

14. The equation $e^x - x - 1 = 0$ has
(A) only one real root (B) at least two real roots
(C) exactly two real roots (D) infinitely many roots.

15. The number of rectangles excluding squares from a rectangle of size 9×6 is
(A) 945 (B) 971
(C) 791 (D) 719

16. The sum of the digits in the unit place of all numbers formed with the help of 3, 4, 5 and 6 taken all at a time is
(A) 180 (B) 108
(C) 118 (D) 181

17. The middle term in the expansion of $(1 + 3x + 3x^2 + x^3)^6$ is
(A) 4th (B) 3rd
(C) 10th (D) none of these

18. If $\frac{1}{1-2x+x^2} = 1 + \alpha_1 x + \alpha_2 x^2 + \cdots$, then the value of a_r is
(A) 2r (B) r + 1
(C) r (D) r – 1

19. The coefficient of x^{17} in the expansion of $(x - 1)(x - 2)(x - 3) \ldots (x - 18)$ is
(A) $\frac{171}{2}$ (B) 342
(C) -171 (D) 684

Space for rough work

20. If z_1, z_2 and z_3 are three distinct complex numbers and a, b, c are three positive real numbers such that $\frac{a}{|z_2 - z_3|} = \frac{b}{|z_3 - z_1|} = \frac{c}{|z_1 - z_2|}$, then $\frac{a^2}{(z_2 - z_3)} + \frac{b^2}{(z_3 - z_1)} + \frac{c^2}{(z_1 - z_2)} =$

(A) 0 (B) abc
(C) 3abc (D) (a + b + c)

21. Let f(z) is a complex polynomial in variable z. When f(z) is divided by z – i, the remainder is i and when divided by z + i, the remainder is 1 + i, then, the remainder, when f(z) is divided by $z^2 + 1$, is

(A) $\frac{1}{2} + i + z$ (B) $\frac{1}{2} + iz + i + \frac{1}{2}$
(C) iz + i + 1 (D) none of these

22. If z_1 , z_2 , z_3 are complex numbers representing the points A, B, C such that $\frac{2}{z_1} = \frac{1}{z_2} + \frac{1}{z_3}$. Then

(A) A, B, C are collinear
(B) circle through points A, B, C has centre at origin O
(C) circle through A, B, C passes through origin
(D) none of these

23. Let α and β be two distinct complex numbers such that $|\alpha| = |\beta|$. If real part of α is positive and imaginary part of β is negative, then the complex number $\frac{\alpha + \beta}{\alpha - \beta}$ may be

(A) real and positive (B) real and negative
(C) purely imaginary (D) zero

24. If x > 1, y > 1, z > 1 are in a G.P., then $\frac{1}{1 + \ln x}, \frac{1}{1 + \ln y}, \frac{1}{1 + \ln z}$ are in

(A) A.P (B) G.P.
(C) H.P. (D) None of these

Space for rough work

25. Sum of the series $S = 1^2 - 2^2 + 3^2 - 4^2 + \ldots - 2002^2 + 2003^2$ is
(A) 2007006 (B) 1005004
(C) 2000506 (D) None of these

26. Let x be the arithmetic mean and y, z be the two geometric means between any two positive numbers. Then the value of $\frac{y^3 + z^3}{xyz}$ is
(A) 2 (B) 3
(C) 1/2 (D) 3/2

27. If $f(x + y) = f(xy)\ \forall\ x, y \in R$, and $f(2003) = 2003$, then $f(-2003)$ equals
(A) 2003 (B) 0
(C) –2003 (D) None of these

28. If $a^2 + 4b^2 = 12ab$, then $\log(a + 2b)$ is
(A) $\frac{1}{2}(\log a + \log b)$ (B) $(\log 4 + \log a + \log b)$
(C) $2\log 2 + \frac{1}{2}(\log a + \log b)$ (D) none of these

29. The formula $\log(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$ to ∞ is valid for
(A) all real x (B) $-1 < x < 1$
(C) $-1 < x \le 1$ (D) $-1 \le x \le 1$

30. If $\tan 3\theta = \cot \theta$, then $\theta =$
(A) $(2n+1)\frac{\pi}{8}$ (B) $(2n+1)\frac{\pi}{4}$
(C) $(n+1)\frac{\pi}{3}$ (D) $n\pi$

Space for rough work